

Highlights

Execution-Aware Model Predictive Value Stacking for Campus EV Charging, Battery Storage, and Wholesale Market Participation

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- Rolling MPC coordinates campus EV charging and BESS dispatch.
- Retail tariffs are coupled with wholesale energy and reserves.
- Gross wholesale revenue can be offset by retail tariff exposure.
- Execution-aware evaluation reveals realized value-stacking limits.

Execution-Aware Model Predictive Value Stacking for Campus EV Charging, Battery Storage, and Wholesale Market Participation

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Abstract

Large workplace electric vehicle (EV) charging facilities can provide flexible demand to distribution and bulk-power systems, but the realized value of this flexibility depends on a coupled operating problem: individual EV sessions must be served reliably while the site manages time-varying retail energy charges, demand charges, battery limits, and wholesale market obligations. This paper develops an execution-aware rolling model predictive control (MPC) framework for a campus EV charging hub co-optimized with a battery energy storage system (BESS). The mixed-integer formulation stacks retail tariff management with day-ahead and real-time wholesale energy and ancillary-service products. A 15-minute rolling MPC updates persistence-based EV forecasts with realized same-day information, solves a fixed 24-hour horizon, implements only the first action, and enforces physical EV availability at execution. A July 2025 campus case study validates exact EV energy reconciliation for both rolling and shrinking-horizon controllers. Rolling MPC improves retail-only net value from \$7,875.04 to \$7,936.10. With wholesale participation, however, the shrinking controller reaches \$8,240.38 while rolling MPC reaches \$7,926.01. Rolling earns more wholesale revenue, but the additional market-oriented dispatch creates larger retail tariff exposure. The results show that gross market revenue is an incomplete measure of EV-BESS value stacking and that execution-aware evaluation is necessary when wholesale products, retail tariffs, and EV service constraints interact behind the meter.

Keywords: Ancillary services, Battery storage, Demand charges, Electric vehicles, Model predictive control, Smart grids, Value stacking

1. Introduction

Workplace and campus EV charging systems are no longer passive electric loads. At sufficient scale, coordinated charging can shift demand across time, reduce demand charges, absorb renewable generation, and contribute operating flexibility to bulk power systems. These opportunities are especially important for large institutional sites, where EV charging, building load, photovoltaic (PV) generation, and BESS operation interact behind a common meter.

The same setting also creates a hard operational problem. EVs arrive unpredictably, remain connected for heterogeneous durations, and require different amounts of energy. A BESS can arbitrage prices and reduce demand charges, but it must preserve state-of-charge (SOC), power limits, and throughput limits. Wholesale market participation adds another layer: energy and ancillary-service awards can generate revenue, but they also impose deliverability and directionality constraints. A policy that appears profitable in a day-ahead schedule may lose value in real time if EV uncertainty creates infeasible charging obligations or if market actions increase retail tariff costs.

This paper studies the value-stacking problem for a large campus EV charging facility with a co-located BESS. Figure 1

summarizes the behind-the-meter system and the coupling between EV charging, building demand, PV, storage, and grid import. The proposed controller combines a day-ahead schedule with a 15-minute real-time rolling MPC. Unlike a conventional shrinking-horizon implementation, the real-time controller always solves a fixed 96-step horizon, maintaining a full 24-hour look-ahead at every interval. The rolling horizon is updated with realized same-day EV information and a persistence forecast for unobserved future intervals. The formulation is designed around a strict execution principle: only the first control action is implemented, and that action must be feasible for EVs physically present at the current interval.

The contributions are as follows.

- A unified EV-BESS value-stacking formulation is developed for retail tariff management and wholesale market participation with day-ahead and real-time energy and ancillary-service products.
- A rolling MPC implementation is introduced with dynamic persistence updates, same-day EV service closure, and first-step physical availability enforcement.
- A market revenue decomposition is provided to attribute wholesale value by asset class, EV flexibility versus BESS, and by function, energy versus capacity.

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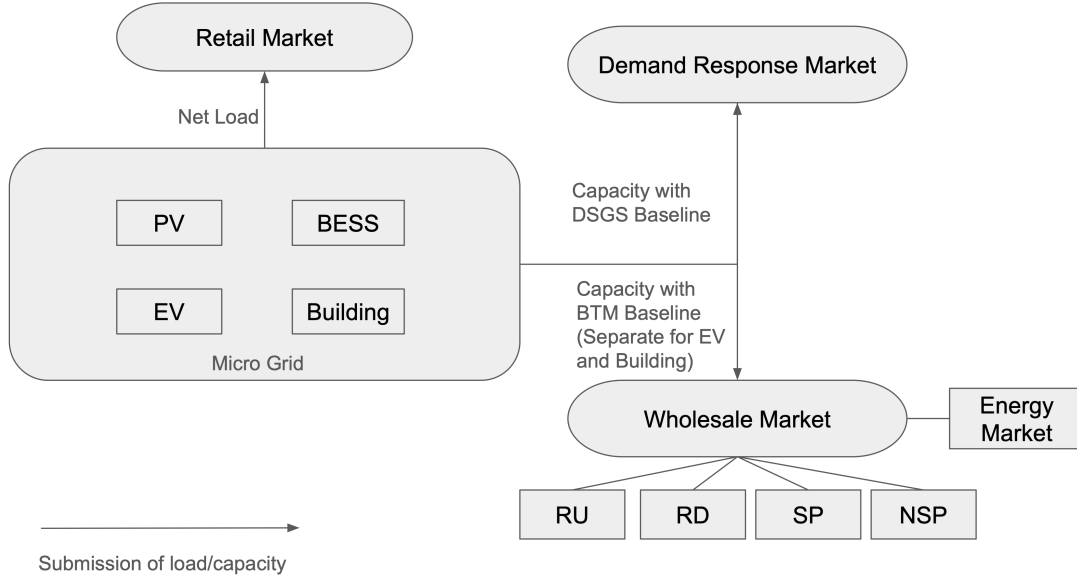


Figure 1: Campus EV-BESS value-stacking system architecture. The optimizer coordinates EV charging, BESS operation, building load, PV generation, retail tariff exposure, and wholesale market participation behind a common grid interconnection.

- A month-long campus case study compares rolling and shrinking-horizon controllers and quantifies when wholesale revenue is offset by retail tariff exposure.

2. Related Work

Coordinated EV charging has been widely studied as a deferrable load scheduling problem, with early work showing how charging control can reduce system peaks and distribution impacts [1–3]. Subsequent reviews and field-oriented studies emphasize that EV flexibility is strongly shaped by arrival uncertainty, dwell-time heterogeneity, user service requirements, and charger power limits [4–6]. These characteristics make EV charging different from an ideal storage resource: flexibility is available only while vehicles are connected, and service quality is judged by delivered session energy rather than by aggregate power tracking alone.

BESS and flexible loads have also been studied as resources for price-responsive operation and ancillary-service provision [7–9]. Storage can supply symmetric power capacity and SOC-backed duration, while EV charging can provide upward flexibility by reducing charging relative to a baseline and downward flexibility by increasing charging. Value stacking across energy arbitrage, demand-charge management, and reserve products is therefore attractive, but it also creates interactions: a dispatch that increases wholesale revenue may raise retail energy charges or create a new demand-charge peak. This retail-wholesale coupling is especially important for large behind-the-meter sites, where market revenue and tariff exposure are settled through different accounting mechanisms but are controlled by the same physical EV-BESS schedule.

MPC is a natural tool for this setting because it repeatedly updates decisions as forecasts and system states change. Prior work has applied predictive control to EV fleets and microgrids under uncertainty, including receding-horizon methods for cost reduction, renewable integration, and reserve provision [10–12]. However, many planning studies report perfect-foresight or full-horizon schedules, while practical operation implements only the next action before forecasts are refreshed. This distinction matters when forecast EVs appear in the look-ahead horizon but are not physically connected at the current interval.

This paper differs from most EV scheduling studies in three ways. First, it evaluates EV charging behind a campus meter where retail demand charges materially affect the profitability of wholesale market actions. Second, it models multiple wholesale products rather than a single reserve or energy-arbitrage product. Third, it focuses on execution consistency in rolling MPC: forecast EVs may influence future look-ahead decisions, but they cannot receive the current executed action before physically arriving, and current-day service obligations cannot be pushed into a forecast tail beyond midnight.

3. System Description and Control Workflow

The site consists of an EV charging fleet, a BESS, building load, and PV generation behind a common grid interconnection. The simulation uses a 15-minute interval, so $\Delta t = 0.25$ h and one operating day has $T = 96$ intervals. Let $\mathcal{T} = \{1, \dots, T\}$ index intervals and $\mathcal{V}$ index EV sessions considered by the optimizer. Positive grid import denotes consumption from the utility. PV generation is represented as non-positive net load.

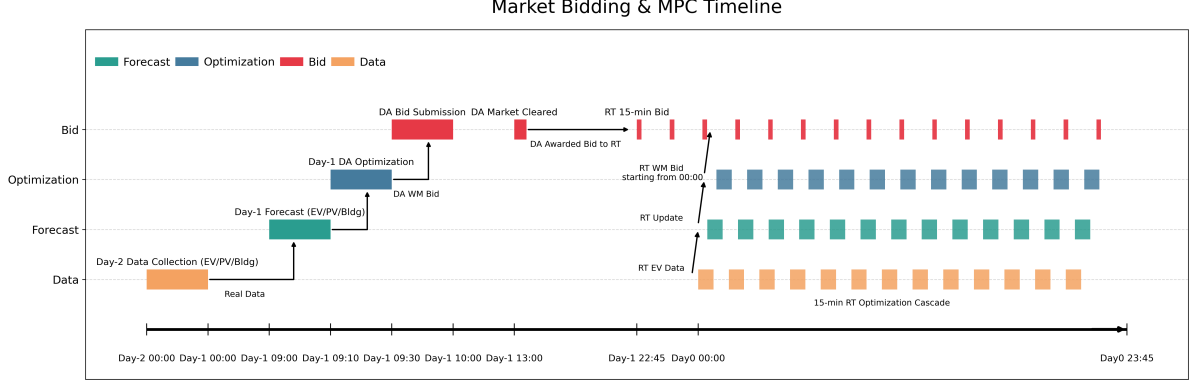


Figure 2: Market and MPC workflow. A day-ahead optimization establishes the initial schedule, while a 15-minute real-time MPC updates EV forecasts, solves a fixed 96-step horizon, executes only the first action, and verifies physical EV energy delivery at the end of each operating day.

The controller is evaluated in two operating modes. In the *retail-only* mode, the optimizer minimizes retail tariff cost net of EV charging revenue and disables wholesale variables. In the *both* mode, the optimizer stacks retail tariff management with wholesale market participation. The wholesale products include day-ahead and real-time energy, regulation up, regulation down, spinning reserve, and non-spinning reserve. The retail cost includes time-of-use (TOU) energy charges, peak-period demand charges (PD), and non-coincident demand charges (NCD). EV service revenue is proportional to delivered EV charging energy.

Figure 2 illustrates the timing of day-ahead bidding, real-time updates, and MPC execution. The key implementation point is that the optimizer can plan over a forecast horizon but only the first interval is executed before forecasts and system states are refreshed.

4. Optimization Model

This section summarizes the EV-BESS value-stacking model using the notation of the implementation and the original formulation. The complete mixed-integer linear programming (MILP) constraints, including binary direction variables, wholesale capacity envelopes, and revenue attribution, are provided in Appendix Appendix A. At real-time step k , the controller optimizes over a horizon $\mathcal{H}_k$ with interval length Δt .

4.1. EV charging model

For EV i , let a_i and d_i denote the arrival and departure intervals, $b_{t,i}^{\text{EV}}$ the connectedness parameter, $p_{t,i}^{\text{EV}}$ the charging power, $e_{t,i}$ the delivered-energy state, and $e_{d,i}$ the required departure energy. The basic service constraints are

$$b_{t,i}^{\text{EV}} = \begin{cases} 1, & a_i \leq t \leq d_i, \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

$$e_{0,i} = 0, \quad (2)$$

$$e_{t,i} = e_{t-1,i} + p_{t,i}^{\text{EV}} \Delta t / \eta_i^{\text{EV}}, \quad \forall t \in \mathcal{H}_k, i \in \mathcal{V}, \quad (3)$$

$$0 \leq p_{t,i}^{\text{EV}} \leq \hat{p}^{\text{EV}} b_{t,i}^{\text{EV}}, \quad \forall t, i, \quad (4)$$

$$e_{d,i} \geq e_{d,i}, \quad \forall i \in \mathcal{V}. \quad (5)$$

The aggregate EV charging power and aggregate available EV charging capability are

$$p_t^{\text{EV}} = \sum_{i \in \mathcal{V}} p_{t,i}^{\text{EV}}, \quad (6)$$

$$p_{t,\max}^{\text{EV}} = N_t^{\text{EV}} \hat{p}^{\text{EV}}. \quad (7)$$

In the rolling implementation, the first real-time action is additionally clipped to physically present EVs, and current-day energy obligations are required to close within the current physical day rather than in the beyond-midnight forecast tail.

4.2. BESS and behind-the-meter balance

Let $p_t^{\text{ch,BESS}}$ and $p_t^{\text{dch,BESS}}$ be BESS charging and discharging powers, C^{BESS} the BESS energy capacity, $p_{\max}^{\text{BESS}}$ the BESS power rating, and γ the round-trip efficiency. The BESS state of charge is

$$\text{soc}_t^{\text{BESS}} = \text{soc}_{t-1}^{\text{BESS}} + \frac{\Delta t}{C^{\text{BESS}}} \left(p_t^{\text{ch,BESS}} \sqrt{\gamma} - \frac{p_t^{\text{dch,BESS}}}{\sqrt{\gamma}} \right), \quad (8)$$

with SOC, power, throughput, and terminal constraints given in Appendix Appendix A. The net BESS power is defined as

$$p_t^{\text{BESS}} = p_t^{\text{ch,BESS}} - p_t^{\text{dch,BESS}}. \quad (9)$$

Following the sign convention in the implementation, $p_t^{\text{PV}} \leq 0$ denotes on-site PV generation and $p_t^{\text{building}} \geq 0$ denotes building demand. The grid import is

$$p_t^{\text{GI}} = p_t^{\text{building}} + p_t^{\text{EV}} + p_t^{\text{BESS}} + p_t^{\text{PV}}. \quad (10)$$

4.3. Retail cost and wholesale revenue

The interval retail cost includes time-of-use energy charges, peak-demand charges, and non-coincident demand charges:

$$\begin{aligned} \text{Cost}_t = & C^{\text{NCD}} [p_t^{\text{GI}} - \overline{p^{\text{GI}}}]_+ + C^{\text{PD}} [z_t^{\text{PD}} p_t^{\text{GI}} - \overline{p_{\text{PD}}^{\text{GI}}}]_+ \\ & + \Delta t C_t^{\text{TOU}} p_t^{\text{GI}}. \end{aligned} \quad (11)$$

The EV charging revenue is

$$R_t^{\text{EV}} = C^{\text{EV}} p_t^{\text{EV}} \Delta t. \quad (12)$$

Let p_t^{DA} and p_t^{RT} be day-ahead and real-time wholesale energy variables. Let $c_t^{k,s}$ be the capacity award for ancillary-service product $k \in \{\text{RU}, \text{RD}, \text{SP}, \text{NSP}\}$ and market stage $s \in \{\text{DA}, \text{RT}\}$. The wholesale revenue follows the original energy-plus-capacity decomposition:

$$R_t^{\text{WM}} = \Delta t \left(C_t^{\text{LMP,DA}} p_t^{\text{DA}} + C_t^{\text{LMP,RT}} p_t^{\text{RT}} \right) + \Delta t \sum_{k \in \mathcal{K}} \sum_{s \in \mathcal{S}} c_t^{k,s} TTP_t^{k,s}, \quad (13)$$

$$TTP_t^{k,s} = C_t^{k,s} + \sigma^k C_t^{\text{LMP,RT}} \alpha_t^k, \quad (14)$$

where $\mathcal{K} = \{\text{RU}, \text{RD}, \text{SP}, \text{NSP}\}$, $\mathcal{S} = \{\text{DA}, \text{RT}\}$, $\sigma^k = 1$ for upward products $\{\text{RU}, \text{SP}, \text{NSP}\}$, and $\sigma^k = -1$ for regulation down. The total rolling-horizon objective is

$$\max \sum_{t \in \mathcal{H}_k} \left(R_t^{\text{EV}} + R_t^{\text{WM}} - \text{Cost}_t - \rho_t^{\text{WM}} \right), \quad (15)$$

where ρ_t^{WM} penalizes deviations from previously committed wholesale schedules. In retail-only mode, all wholesale variables and R_t^{WM} are fixed to zero.

5. Execution-Aware Rolling MPC

The day-ahead stage solves a full-day optimization using forecast EV, building, PV, and price information. The real-time stage implements a rolling MPC loop. At every 15-minute step k , the optimizer receives a fixed 96-interval horizon. The horizon contains the remaining portion of the current day and a beyond-midnight tail. This is different from a shrinking-horizon controller, whose horizon length decreases from 96 to 1 intervals during the day.

5.1. Dynamic Persistence Forecast Update

The persistence forecast is initialized from historical data. During the current day, its prefix is overwritten with observed EV arrivals and realized availability. Conceptually, at real-time step k ,

$$\begin{aligned} \underbrace{1:k}_{\text{observed}} &\leftarrow \text{real day-0 EV data,} \\ \underbrace{k+1:96}_{\text{unobserved}} &\leftarrow \text{persistence forecast,} \\ \underbrace{97:k+96}_{\text{tail}} &\leftarrow \text{executed day-0 EV load.} \end{aligned} \quad (16)$$

The tail is intentionally not padded with zeros. Zero padding creates an artificially low future load after midnight and can distort BESS and wholesale decisions near the end of the day.

5.2. Execution-Aware Corrections

Two implementation details are essential for physical feasibility. First, before the real-time solve, the first row of the EV upper-bound matrix is clipped to actual current availability:

$$p_{0,i}^{\text{EV,MPC}} \leftarrow \min\{p_{0,i}^{\text{EV,forecast}}, p_{0,i}^{\text{EV,real}}\}. \quad (17)$$

Thus, forecast EVs may affect future look-ahead decisions but cannot receive energy in the current executed interval unless they have physically arrived. Second, the current-day service closure constraint ensures that the optimizer cannot move required day-0 EV energy into the beyond-midnight forecast tail. At the end of every day, the implementation checks

$$\Delta t \sum_{i=1}^{96} \sum_i p_{t,i}^{\text{EV,executed}} = \sum_i e_{d,i,0}^{\text{true}}. \quad (18)$$

The simulation stops if this equality fails outside numerical tolerance.

5.3. Implementation Notes from the Simulation Code

The implementation uses two related optimization models. The day-ahead model solves a 96-interval daily problem and initializes wholesale schedules, BESS state, monthly demand-charge thresholds, and baseline EV charging profiles. The real-time model then reuses the same CVXPY problem structure with updated parameters at every 15-minute interval. In the wholesale-enabled mode, deviation variables penalize departures from prior day-ahead wholesale commitments for energy and ancillary products, while in retail-only mode all wholesale variables, EV wholesale flexibility variables, and BESS wholesale channels are fixed to zero. This distinction is important for interpreting runtime and economics: the retail-only case is evaluated using the same model family for comparability, rather than a separately simplified retail-only optimizer.

6. Case Study

The case study evaluates a UC San Diego campus EV charging site over July 2025. The simulation uses 15-minute EV session data, campus load and PV profiles, retail tariff components, and wholesale market prices. The rolling-MPC run uses persistence forecasts for EV energy and EV count, perfect arrival-time information, and a fixed 96-step real-time horizon. The benchmark shrinking controller uses the same data, forecast setting, market configuration, solver thread count, MIP gap, and real-time solve time limit, but its real-time horizon shortens as the operating day progresses. For each controller, two operating modes are evaluated: retail-only and combined retail-plus-wholesale operation. The implementation evaluates controllable EV-BESS import relative to the precomputed campus baseline; equivalently, the full behind-the-meter balance in Eq. (10) is recovered by adding building load and PV terms to this controllable import.

Table 2 summarizes the code-confirmed parameters used in the simulations. The optimization notebooks implement the day-ahead problem and the real-time MPC problem in CVXPY and solve the mixed-integer programs with Gurobi. The real-time implementation uses CVXPY parameters for repeated horizon updates, which keeps the problem structure fixed while refreshing EV availability, remaining energy, tariff thresholds, BESS SOC, market prices, and prior wholesale commitments at each interval.

Table 1: Execution-aware rolling MPC procedure.

Step	Operation
1	Initialize the day-ahead schedule and persistence EV forecast.
2	At interval k , update newly arrived EVs using observed session data.
3	Construct a 96-step rolling forecast using Eq. (16).
4	Clip first-step EV availability using Eq. (17).
5	Solve the EV-BESS value-stacking optimization in Eq. (15).
6	Execute only the first interval EV and BESS actions.
7	Update EV remaining energy and BESS SOC from executed actions.
8	At the end of the day, verify Eq. (18).

Table 2: Case-study and implementation parameters confirmed from the simulation notebooks.

Item	Value or implementation setting
Simulation period	July 2025, 31 operating days
Time step	15 min, $\Delta t = 0.25$ h; 96 intervals per day
Rolling real-time horizon	Fixed 96 intervals at every RT step
Shrinking benchmark horizon	Remaining intervals of current operating day
BESS power rating	$P_{\max}^{\text{BESS}} = 250$ kW
BESS energy capacity	$C^{\text{BESS}} = 332$ kWh
BESS SOC range	$0.05 \leq \text{soc}_t^{\text{BESS}} \leq 0.95$
BESS initial/target SOC	0.5 at the beginning of the first simulated day; terminal target on the final simulated day
BESS efficiency parameter	$\gamma = 0.9$ in the SOC update, applied through $\sqrt{\gamma}$ terms
BESS throughput limit	Daily equivalent full-cycle bound: $\Delta t \sum_i (p_i^{\text{ch,BESS}} + p_i^{\text{dch,BESS}}) \leq 2C^{\text{BESS}}(\text{SOC}_{\max} - \text{SOC}_{\min})$
EV charging service price	\$0.40/kWh
Maximum EV charging power	6.6 kW per connected EV session
Retail tariff season	Summer tariff setting
TOU windows	Super-off-peak 00:00–06:00; off-peak 06:00–16:00 and 21:00–24:00; on-peak 16:00–21:00
Demand-charge rates	PD: \$3.05/kW; NCD: \$15.38/kW
Ancillary deployment factors	$\alpha_{\text{RU}} = \alpha_{\text{RD}} = 0.7$, $\alpha_{\text{SP}} = \alpha_{\text{NSP}} = 0.2$
AS duration requirement	30 min, implemented as $2\Delta t$ energy headroom/footroom
Optimization software	CVXPY with Gurobi
Solver settings	MIP gap 10^{-3} ; 10 solver threads; RT time limit 0.5 s per interval
Wholesale modes	Retail-only disables all wholesale variables; Both enables DA/RT energy and RU/RD/SP/NSP products

Both controllers pass the same strict EV energy reconciliation test. Across the 31 simulated days, the true EV session energy is 54,869.506 kWh, and the executed base-case dispatch equals this value within numerical tolerance for all four controller-mode combinations. This validation is important because economic comparisons are meaningful only when the EV service obligation is held fixed.

7. Results

7.1. Monthly Value Stack

Table 3 summarizes the monthly value stacks for rolling and shrinking MPC. All cases deliver the same EV charging revenue because the base-case EV service energy is reconciled exactly. In the rolling controller, wholesale participation adds \$3,735.76 in market revenue but increases TOU exposure by \$2,303.58, peak-demand-charge exposure by \$189.32, and non-coincident-demand-charge exposure by \$1,252.93 relative to rolling

retail-only operation. Consequently, the rolling wholesale-enabled case is \$10.09 lower than rolling retail-only operation. In contrast, the shrinking controller gains \$365.34 from wholesale participation relative to shrinking retail-only operation.

Figure 3 shows the monthly rolling value-stack components. The figure highlights that wholesale revenue is economically meaningful in gross terms, but its net effect depends on the additional retail tariff exposure created by market-oriented dispatch.

7.2. Rolling Versus Shrinking Benchmark

Table 4 and Fig. 4 compare rolling and shrinking MPC by operating mode. Rolling improves the retail-only case by \$61.06, while shrinking outperforms rolling by \$314.37 in the wholesale-enabled case. The latter result is not a physical feasibility error: both controllers dispatch exactly the same total EV service energy and use the same input data and solver settings. Instead, rolling earns \$222.64 more wholesale revenue than shrinking in

Table 3: Monthly value stack for July 2025. Positive entries are revenue; negative entries are costs.

Controller and case	Total	WM	TOU	PD	NCD	EV
Shrinking retail-only	7875.04	0.00	-9039.14	-281.14	-4752.45	21947.80
Shrinking both	8240.38	3513.12	-10490.10	-300.81	-6429.66	21947.80
Rolling retail-only	7936.10	0.00	-8546.65	-88.91	-5376.12	21947.80
Rolling both	7926.01	3735.76	-10850.23	-278.23	-6629.05	21947.80

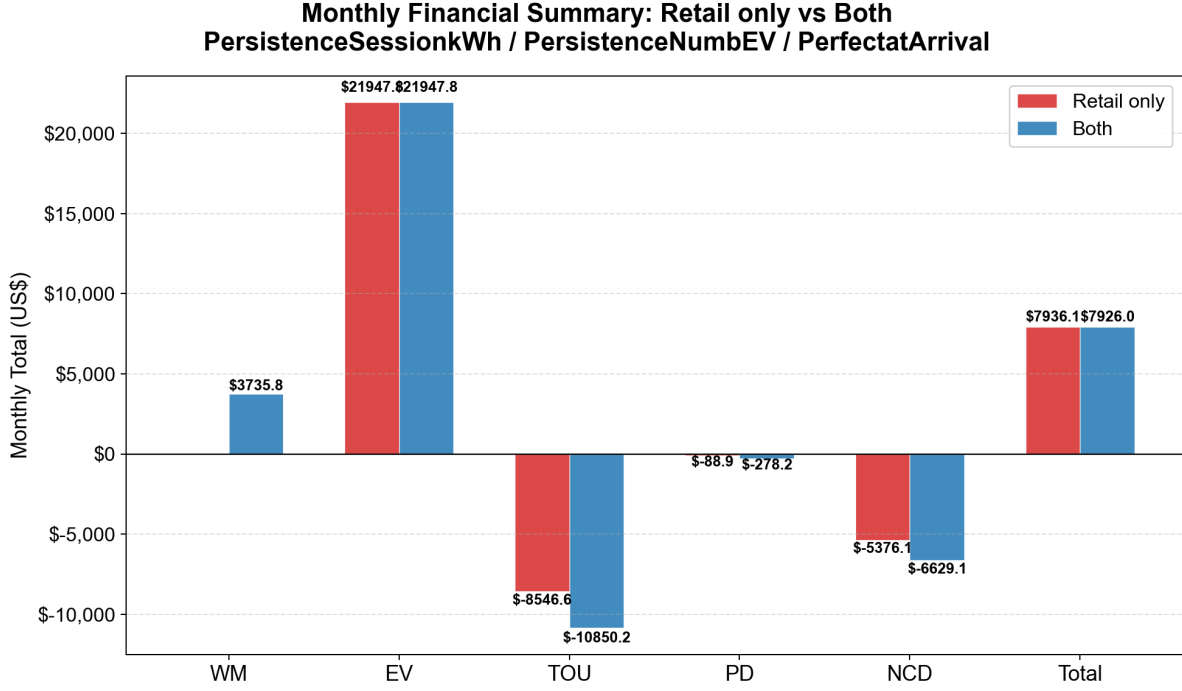


Figure 3: Monthly value-stack comparison for rolling MPC. Wholesale participation increases gross market revenue but can shift grid import in ways that increase tariff costs.

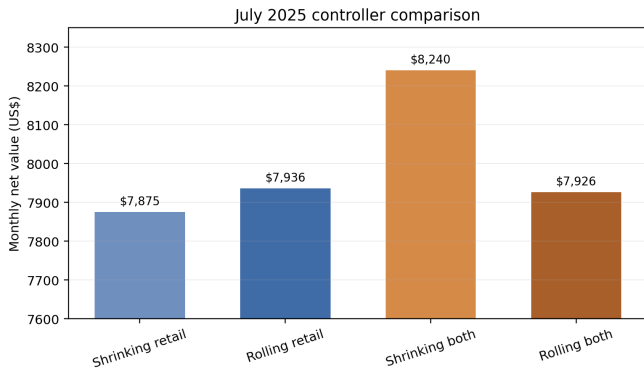


Figure 4: Net monthly value comparison between shrinking and rolling MPC. Rolling improves the retail-only case, while shrinking achieves the highest value when wholesale participation is enabled.

the Both case, but the additional revenue is more than offset by higher TOU and NCD costs.

7.3. Wholesale Revenue Attribution

Table 5 decomposes the rolling wholesale revenue. The EV fleet contributes \$2,519.82, approximately 67.5% of wholesale value, while the BESS contributes \$1,215.79. Energy-related wholesale revenue dominates capacity revenue in this month, with \$3,025.07 from energy and expected deployment effects versus \$709.94 from capacity payments. This decomposition shows that the EV charging fleet provides most of the wholesale flexibility value, but that value does not necessarily translate into higher total net profit after tariff impacts are included. The attribution formulas used for this decomposition are given in Appendix Appendix B.

7.4. Daily Variability

The monthly totals conceal large day-to-day variation. Under rolling MPC with wholesale participation, daily total net value ranges from -\$4,128.50 to \$802.37. The most adverse day is July 1, when high demand-charge exposure dominates the value stack. Figure 6 shows the daily value-stack comparison, while Fig. 7 provides a broader daily financial view. Together, these figures illustrate why a value-stacking controller

Table 4: Controller comparison by operating mode.

Mode	Rolling	Shrinking	Rolling – shrinking (US\$)
Retail-only	7936.10	7875.04	61.06
Both	7926.01	8240.38	-314.37

Table 5: Confirmed rolling-MPC wholesale revenue decomposition for the Both case.

Component	Monthly value (US\$)	Share
BESS total	1215.79	32.6%
EV total	2519.82	67.5%
Energy revenue	3025.07	81.0%
Capacity revenue	709.94	19.0%

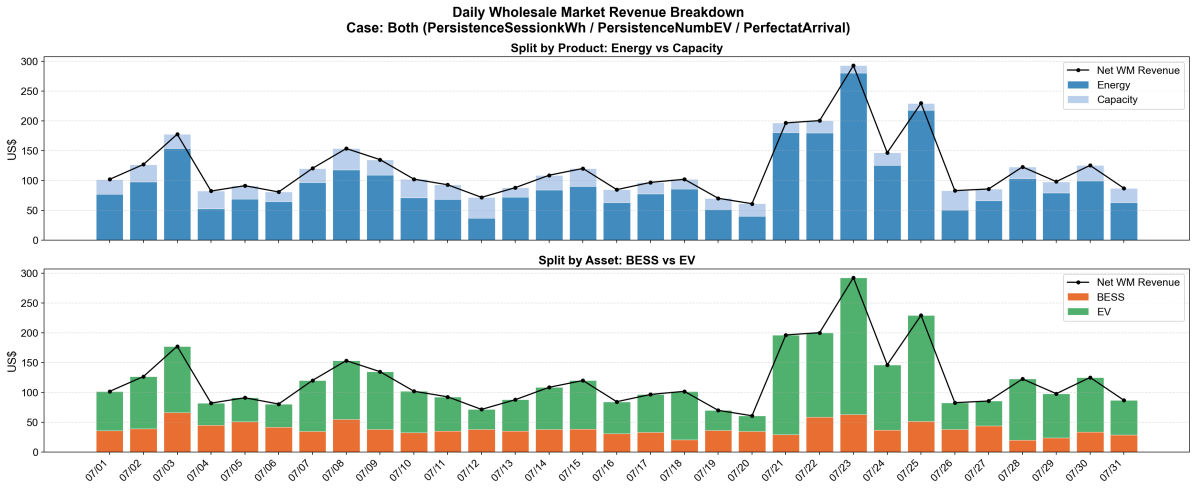


Figure 5: Daily wholesale market revenue decomposition under rolling MPC. EV flexibility provides most of the monthly wholesale value, while BESS revenue remains material because it supplies symmetric power capability and SOC-backed duration.

must optimize retail and wholesale terms jointly: a market-positive action can still be net-negative if it creates a new demand-charge peak.

8. Discussion

The results show that wholesale market participation must be evaluated as a complete value stack, not as gross market revenue alone. In the rolling controller, the Both case earns substantial wholesale revenue, but the retail-only case slightly outperforms it in total net value. In the shrinking controller, wholesale participation is profitable relative to retail-only operation and achieves the highest monthly value among the four tested cases. The contrast is useful: the same data, tariff structure, solver settings, and market products can produce different economic rankings solely because the real-time horizon changes the intertemporal allocation of EV and BESS flexibility.

The comparison also clarifies the role of longer look-ahead. A fixed 24-hour horizon is not a guarantee of higher realized profit. It provides additional forecast context, but that context is only valuable if the forecast tail and objective accurately rep-

resent future economic consequences. With persistence forecasts, a longer horizon can anticipate future load pressure and preserve flexibility, but it can also schedule actions that increase retail exposure. The tested month shows precisely this tension: rolling MPC captures more wholesale revenue in the Both case, yet the additional tariff costs dominate. For campus aggregators, this means that a market-bidding controller should be assessed using realized behind-the-meter settlement, not only awarded market revenue or DA/RT price spreads.

The result also identifies a design tradeoff for value-stacking algorithms. The EV fleet contributes most of the wholesale revenue because baseline-relative charging flexibility is large compared with the BESS rating during many intervals. However, EV flexibility is not storage: using it for market products changes when service energy is delivered, and those timing changes are settled under the retail tariff. The BESS supplies SOC-backed capability and can buffer some of this timing risk, but its 250 kW/332 kWh rating is small relative to the aggregate charging fleet during high-utilization periods. Consequently, reserve and energy awards that are feasible in a power-capability sense may still be unattractive after demand charges

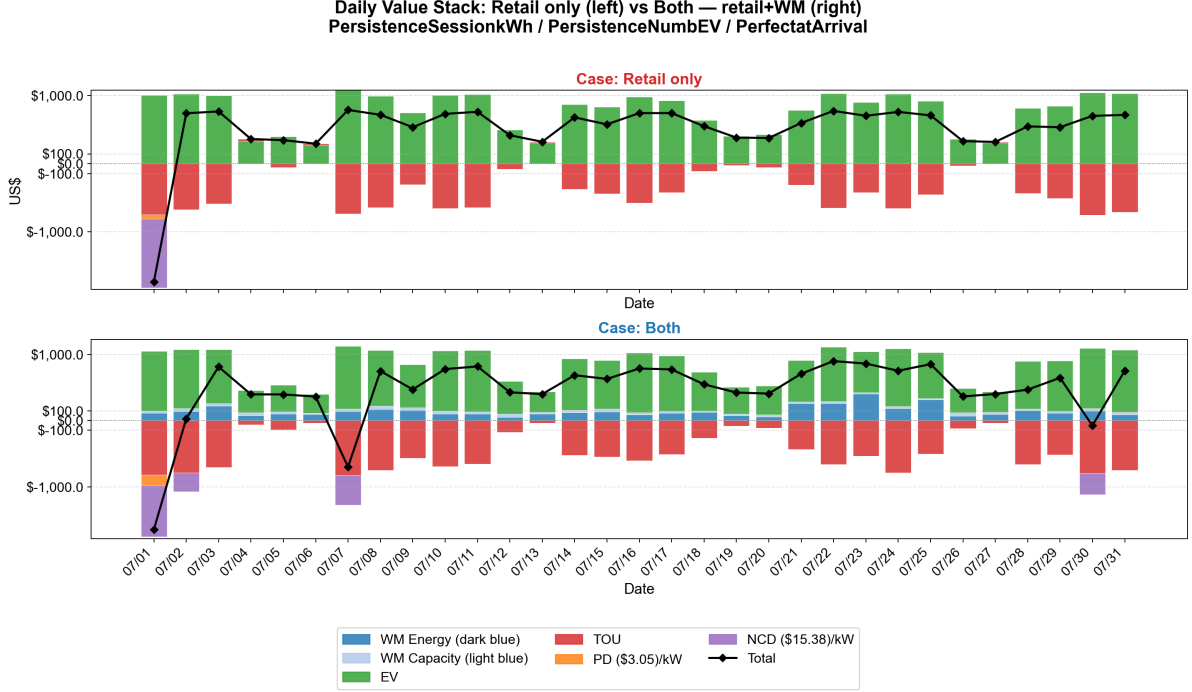


Figure 6: Daily value-stack comparison under rolling MPC. Demand charges create sparse but high-impact cost events that can dominate daily economics.

are included.

The results also emphasize the importance of execution-aware modeling. Without first-step availability clipping, a rolling optimizer can allocate immediate energy to forecast EVs that have not arrived. Without the same-day service constraint, the optimizer can satisfy current-day EV requirements in the beyond-midnight persistence tail. Both errors can make the optimization appear feasible while causing physical dispatch mismatch. The proposed implementation avoids these artifacts and verifies exact same-day EV energy delivery. This is a methodological contribution in addition to an implementation detail: it changes the reported economic comparison from an idealized plan to a physically executed dispatch.

Several limitations remain. The July 2025 study uses persistence-based EV energy and count forecasts with perfect arrival-time information, so the results should be interpreted as an execution-aware deterministic benchmark rather than a complete uncertainty analysis. The retail-only mode is solved within the same model family as the wholesale-enabled case for comparability; a reduced retail-only MILP could further reduce runtime. Finally, the reported month is a single tariff and market condition. Future work should evaluate additional seasons, forecast-error models, BESS sizes, solve-time limits, and market-product sub-sets.

9. Conclusion

This paper presented an execution-aware rolling MPC for EV-BESS value stacking under retail tariffs and wholesale market participation. The controller maintains a fixed 24-hour horizon, dynamically updates persistence forecasts, enforces phys-

ical first-step EV availability, and prevents same-day EV service obligations from being shifted beyond midnight. A July 2025 campus case study confirms strict EV energy reconciliation for both rolling and shrinking-horizon controllers. Rolling MPC achieves the higher retail-only value, \$7,936.10 versus \$7,875.04 for shrinking MPC. In contrast, shrinking MPC achieves the higher wholesale-enabled value, \$8,240.38 versus \$7,926.01 for rolling MPC. The result is not a failure of rolling MPC or value stacking; rather, it demonstrates that longer look-ahead and additional market revenue can be outweighed by retail tariff exposure.

Future work will evaluate stochastic or distributionally robust EV forecasts, tune the persistence tail used in fixed-horizon rolling MPC, and develop a reduced retail-only optimization model that omits disabled wholesale variables for faster long-horizon simulation. Another important extension is to quantify sensitivity to real-time solve time limits, because suboptimal mixed-integer solutions can influence the tradeoff between market revenue and tariff cost.

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Appendix A. Complete MILP Formulation

This appendix collects the detailed constraints used to make the compact formulation in Section 4 implementable. The notation follows the main text. The complete implementation includes forecast-dependent indexing and solver-specific book-

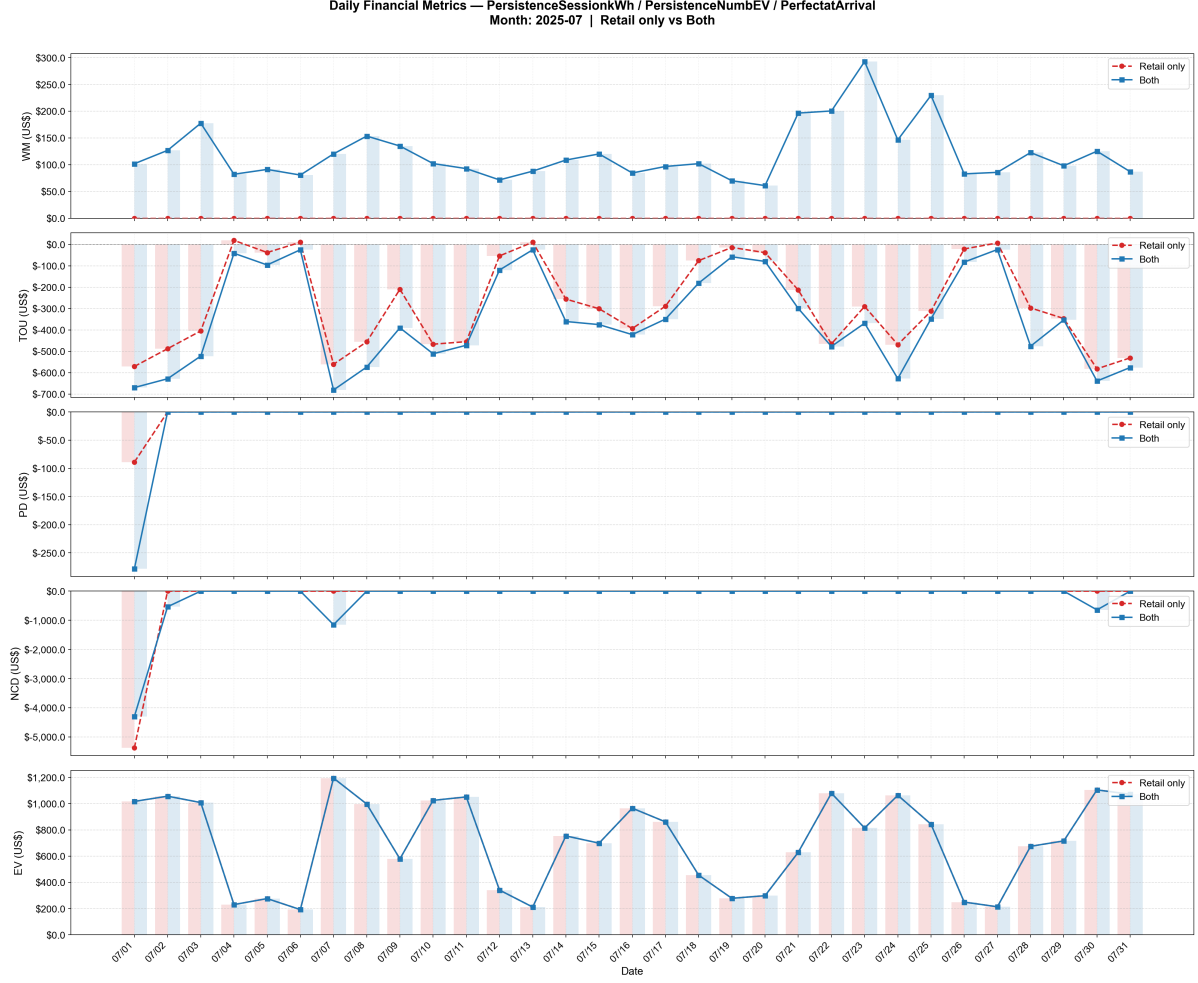


Figure 7: Daily financial metrics for the rolling MPC case. This diagnostic figure complements the value-stack view by showing how daily revenue and retail cost components vary across the simulated month.

keeping; the equations below state the physical and market constraints that define the controller.

Appendix A.1. EV Arrival, Departure, and Energy State

For EV i , let a_i and d_i denote the arrival and departure intervals. The connectedness parameter is

$$b_{t,i}^{\text{EV}} = \begin{cases} 1, & a_i \leq t \leq d_i, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{A.1})$$

Let $e_{t,i}$ be the cumulative delivered energy state. The boundary and dynamics constraints are

$$e_{0,i} = 0, \quad (\text{A.2})$$

$$e_{a_i,i} = e_{a_i}, \quad (\text{A.3})$$

$$e_{d_i,i} = e_{d_i}, \quad (\text{A.4})$$

$$e_{t,i} = e_{t-1,i} + p_{t,i}^{\text{EV}} \Delta t / \eta_i^{\text{EV}}, \quad \forall t, i, \quad (\text{A.5})$$

$$0 \leq e_{t,i} \leq e_{d_i}, \quad \forall t, i, \quad (\text{A.6})$$

$$0 \leq p_{t,i}^{\text{EV}} \leq \hat{p}^{\text{EV}} b_{t,i}^{\text{EV}}, \quad \forall t, i. \quad (\text{A.7})$$

The aggregate EV power and connected charging capability are

$$p_t^{\text{EV}} = \sum_{i \in \mathcal{V}} p_{t,i}^{\text{EV}}, \quad (\text{A.8})$$

$$p_{t,\max}^{\text{EV}} = N_t^{\text{EV}} \hat{p}^{\text{EV}}. \quad (\text{A.9})$$

Appendix A.2. BESS SOC, Terminal Target, and Wholesale Split

The BESS SOC dynamics are

$$\text{soc}_t^{\text{BESS}} = \text{soc}_{t-1}^{\text{BESS}} + \frac{\Delta t}{C^{\text{BESS}}} \left(p_t^{\text{ch,BESS}} \sqrt{\gamma} - \frac{p_t^{\text{dch,BESS}}}{\sqrt{\gamma}} \right). \quad (\text{A.10})$$

The BESS satisfies

$$\text{SOC}_{\min}^{\text{BESS}} \leq \text{soc}_t^{\text{BESS}} \leq \text{SOC}_{\max}^{\text{BESS}}, \quad (\text{A.11})$$

$$\text{soc}_0^{\text{BESS}} = \text{soc}_{\text{init}}^{\text{BESS}}, \quad (\text{A.12})$$

$$\text{soc}_T^{\text{BESS}} = \text{soc}_{\text{tar}}^{\text{BESS}} \quad \text{on the final simulated day.} \quad (\text{A.13})$$

$$p_t^{\text{BESS}} = p_t^{\text{ch,BESS}} - p_t^{\text{dch,BESS}}, \quad (\text{A.14})$$

$$-P_{\max}^{\text{BESS}} \leq p_t^{\text{BESS}} \leq P_{\max}^{\text{BESS}}, \quad (\text{A.15})$$

$$\Delta t \sum_{t=1}^T (p_t^{\text{ch,BESS}} + p_t^{\text{dch,BESS}}) \leq 2C^{\text{BESS}} (\text{SOC}_{\max}^{\text{BESS}} - \text{SOC}_{\min}^{\text{BESS}}). \quad (\text{A.16})$$

To distinguish wholesale from non-wholesale operation,

$$p_t^{\text{ch,BESS}} = p_t^{\text{ch,BESS,WM}} + p_t^{\text{ch,BESS,NWM}}, \quad (\text{A.17})$$

$$p_t^{\text{dch,BESS}} = p_t^{\text{dch,BESS,WM}} + p_t^{\text{dch,BESS,NWM}}. \quad (\text{A.18})$$

Appendix A.3. Wholesale Energy and Ancillary-Service Capacity Envelopes

Wholesale energy variables p_t^{DA} and p_t^{RT} are free variables, while ancillary-service awards $c_t^{q,s}$ are nonnegative. Binary variables enforce mutually exclusive net charging/discharging directions. For a BESS-only capability envelope,

$$\hat{p}_t^{\text{up}} = \hat{p}_t^{\text{down}} = P_{\max}^{\text{BESS}}. \quad (\text{A.19})$$

For the combined BESS+EV resource,

$$\hat{p}_t^{\text{up}} = P_{\max}^{\text{BESS}} + B_t^{\text{EV}}, \quad (\text{A.20})$$

$$\hat{p}_t^{\text{down}} = P_{\max}^{\text{BESS}} + P_{t,\max}^{\text{EV}} - B_t^{\text{EV}}. \quad (\text{A.21})$$

To state the MILP without nonlinear positive-part notation, introduce nonnegative variables $p_t^{\text{DA},+}$, $p_t^{\text{DA},-}$, $p_t^{\text{RT},+}$, and $p_t^{\text{RT},-}$ such that

$$p_t^{\text{DA}} = p_t^{\text{DA},+} - p_t^{\text{DA},-}, \quad p_t^{\text{RT}} = p_t^{\text{RT},+} - p_t^{\text{RT},-}, \quad (\text{A.22})$$

$$p_t^{\text{DA},+}, p_t^{\text{DA},-}, p_t^{\text{RT},+}, p_t^{\text{RT},-} \geq 0. \quad (\text{A.23})$$

The energy and ancillary-service awards are constrained by

$$p_t^{\text{DA},+} + p_t^{\text{RT},+} + c_t^{\text{RU,DA}} + c_t^{\text{RU,RT}} + c_t^{\text{SP,DA}} + c_t^{\text{SP,RT}} + c_t^{\text{NSP,DA}} + c_t^{\text{NSP,RT}} \leq \hat{p}_t^{\text{up}}, \quad (\text{A.24})$$

$$p_t^{\text{DA},-} + p_t^{\text{RT},-} + c_t^{\text{RD,DA}} + c_t^{\text{RD,RT}} \leq \hat{p}_t^{\text{down}}, \quad (\text{A.25})$$

$$-\hat{p}_t^{\text{down}} \leq p_t^{\text{DA}} \leq \hat{p}_t^{\text{up}}, \quad (\text{A.26})$$

$$-\hat{p}_t^{\text{down}} \leq p_t^{\text{RT}} \leq \hat{p}_t^{\text{up}}. \quad (\text{A.27})$$

The CVXPY implementation uses convex positive-part expressions that are converted to equivalent solver constraints; the equations above give the explicit linearized representation.

Appendix A.4. Charge/Discharge Direction Binaries and EV Big-M Decomposition

$\forall t$ The BESS, EV fleet, and net wholesale resource cannot simultaneously charge and discharge:

$$b_t^{\text{ch,BESS}} + b_t^{\text{dch,BESS}} \leq 1, \quad (\text{A.28})$$

$$b_t^{\text{ch,EV}} + b_t^{\text{dch,EV}} \leq 1, \quad (\text{A.29})$$

$$b_t^{\text{ch,net}} + b_t^{\text{dch,net}} \leq 1. \quad (\text{A.30})$$

$\forall t$ The BESS direction variables impose

$$p_t^{\text{dch,BESS}} \leq P_{\max}^{\text{BESS}} b_t^{\text{dch,BESS}}, \quad (\text{A.31})$$

$$p_t^{\text{ch,BESS}} \leq P_{\max}^{\text{BESS}} b_t^{\text{ch,BESS}}. \quad (\text{A.32})$$

The EV fleet can provide upward flexibility by reducing charging relative to baseline and downward flexibility by increasing charging. Let $M_t^{\text{EV}} = 2P_{t,\max}^{\text{EV}}$. Then

$$p_t^{\text{EV}} - B_t^{\text{EV}} = p_t^{\text{ch,EV}} - p_t^{\text{dch,EV}}, \quad (\text{A.33})$$

$$p_t^{\text{ch,EV}} \leq M_t^{\text{EV}} b_t^{\text{ch,EV}}, \quad (\text{A.34})$$

$$p_t^{\text{dch,EV}} \leq M_t^{\text{EV}} b_t^{\text{dch,EV}}. \quad (\text{A.35})$$

The net wholesale charging/discharging decomposition is

$$p_t^{\text{ch,net}} - p_t^{\text{dch,net}} = (p_t^{\text{ch,BESS,WM}} - p_t^{\text{dch,BESS,WM}}) + (p_t^{\text{ch,EV}} - p_t^{\text{dch,EV}}), \quad (\text{A.36})$$

$$p_t^{\text{ch,net}} \leq \hat{p}_t^{\text{down}} b_t^{\text{ch,net}}, \quad (\text{A.37})$$

$$p_t^{\text{dch,net}} \leq \hat{p}_t^{\text{up}} b_t^{\text{dch,net}}. \quad (\text{A.38})$$

Appendix A.5. Ancillary-Service Duration Constraints

Let

$$\begin{aligned} c_t^{\text{RU,tot}} &= c_t^{\text{RU,DA}} + c_t^{\text{RU,RT}}, & c_t^{\text{RD,tot}} &= c_t^{\text{RD,DA}} + c_t^{\text{RD,RT}}, \\ c_t^{\text{SP,tot}} &= c_t^{\text{SP,DA}} + c_t^{\text{SP,RT}}, & c_t^{\text{NSP,tot}} &= c_t^{\text{NSP,DA}} + c_t^{\text{NSP,RT}}. \end{aligned} \quad (\text{A.39})$$

For ancillary-service duration $T_{\text{DU}} = 2\Delta t$ in the simulations, the BESS must reserve sufficient energy headroom and footroom:

$$\text{soc}_t^{\text{BESS}} - \text{SOC}_{\min}^{\text{BESS}} \geq \frac{T_{\text{DU}}}{C^{\text{BESS}} \sqrt{\gamma}} (c_t^{\text{RU,tot}} + c_t^{\text{SP,tot}} + c_t^{\text{NSP,tot}}), \quad (\text{A.40})$$

$$\text{SOC}_{\max}^{\text{BESS}} - \text{soc}_t^{\text{BESS}} \geq \frac{T_{\text{DU}} \sqrt{\gamma}}{C^{\text{BESS}}} c_t^{\text{RD,tot}}. \quad (\text{A.41})$$

Appendix B. Wholesale Revenue Decomposition and EV/BESS Attribution

The total wholesale revenue can be decomposed into energy and capacity components:

$$R_t^{\text{Energy}} = \Delta t (C_t^{\text{LMP,DA}} p_t^{\text{DA}} + C_t^{\text{LMP,RT}} p_t^{\text{RT}}), \quad (\text{B.1})$$

$$R_t^{\text{Cap,Up}} = \Delta t \sum_{s \in \{\text{DA,RT}\}} (c_t^{\text{RU},s} \Theta_t^{\text{RU},s} + c_t^{\text{SP},s} \Theta_t^{\text{SP},s} + c_t^{\text{NSP},s} \Theta_t^{\text{NSP},s}), \quad (\text{B.2})$$

$$R_t^{\text{Cap,Down}} = \Delta t \sum_{s \in \{\text{DA,RT}\}} c_t^{\text{RD},s} \Theta_t^{\text{RD},s}. \quad (\text{B.3})$$

The BESS share of upward and downward capacity is approximated by available physical capacity shares:

$$\omega_t^{\text{Cap,Up,BESS}} = \frac{P_{\max}^{\text{BESS}}}{P_{\max}^{\text{BESS}} + B_t^{\text{EV}}}, \quad (\text{B.4})$$

$$\omega_t^{\text{Cap,Dn,BESS}} = \frac{P_{\max}^{\text{BESS}}}{P_{\max}^{\text{BESS}} + P_{t,\max}^{\text{EV}} - B_t^{\text{EV}}}. \quad (\text{B.5})$$

The BESS share of energy revenue is an ex-post accounting metric rather than an optimization constraint. It is based on the BESS net wholesale contribution and clipped to $[0, 1]$ after dispatch is computed:

$$\omega_t^{\text{Energy,BESS}} = \text{clip}_{[0,1]} \left(\frac{P_t^{\text{dch,BESS}} - P_t^{\text{ch,BESS}}}{P_t^{\text{dch,BESS}} - P_t^{\text{ch,BESS}} + P_t^{\text{dch,EV}} - P_t^{\text{ch,EV}} + \epsilon} \right), \quad (\text{B.6})$$

where ϵ is a small denominator guard used only in post-processing. The asset-level revenue attribution is

$$R_t^{\text{BESS}} = \omega_t^{\text{Energy,BESS}} R_t^{\text{Energy}} + \omega_t^{\text{Cap,Up,BESS}} R_t^{\text{Cap,Up}} + \omega_t^{\text{Cap,Dn,BESS}} R_t^{\text{Cap,Dn}}, \quad (\text{B.7})$$

$$R_t^{\text{EV}} = (1 - \omega_t^{\text{Energy,BESS}}) R_t^{\text{Energy}} + (1 - \omega_t^{\text{Cap,Up,BESS}}) R_t^{\text{Cap,Up}} + (1 - \omega_t^{\text{Cap,Dn,BESS}}) R_t^{\text{Cap,Dn}}, \quad (\text{B.8})$$

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